Week 10

Book exercise 7.2)

Suppose that we have some $\underline{w}_0$ and b_0 that define a maximum margin classifier, i.e., satisfies (7.5) while solving (7.3).

Now, by assumption, it must hold that

$$t_n\left((\gamma \underline{w}_0)^T \phi(\underline{x}_n) + (\gamma b_0)\right) \geq \gamma$$

for $n=1,\dots,N$ and $\gamma > 0$.

However the perpendicular distance of a point x to/from the hyperplane remains the same, even when we scale w_0 and b_0 with γ. - See figure 4.1 in the book.

Book exercise 7.7)

We shall take (7.57)-(7.60) as given. Substituting onto (7.56) yields

$$L \overset{(7.56)}{=} C\sum_{n=1}^{N}(\xi_n+\hat{\xi}_n)+\frac{1}{2}\|\underline{w}\|^2-\sum_{n=1}^{N}(\mu_n\xi_n+\hat{\mu}_n\hat{\xi}_n)$$
$$-\sum_{n=1}^{N}a_n(\varepsilon+\xi_n+y_n-t_n)-\sum_{n=1}^{N}\hat{a}_n(\varepsilon+\hat{\xi}_n-y_n+t_n)$$

$$=C\sum_{n=1}^{N}(\xi_n+\hat{\xi}_n)+\frac{1}{2}\|\underline{w}\|^2$$
$$-\sum_{n=1}^{N}(a_n+\mu_n)\xi_n-\sum_{n=1}^{N}(\hat{a}_n+\hat{\mu}_n)\hat{\xi}_n$$
$$-\sum_{n=1}^{N}a_n(\varepsilon+y_n-t_n)-\sum_{n=1}^{N}\hat{a}_n(\varepsilon-y_n+t_n)$$

(7.59) and (7.60)

$$=C\sum_{n=1}^{N}(\xi_n+\hat{\xi}_n)+\frac{1}{2}\|\underline{w}\|^2$$
$$-\sum_{n=1}^{N}C\xi_n-\sum_{n=1}^{N}C\hat{\xi}_n$$
$$-\sum_{n=1}^{N}(a_n+\hat{a}_n)\varepsilon-\sum_{n=1}^{N}a_n(y_n-t_n)-\sum_{n=1}^{N}\hat{a}_n(-y_n+t_n)$$

$$=\frac{1}{2}\|\underline{w}\|^2-\sum_{n=1}^{N}(a_n+\hat{a}_n)\varepsilon-\sum_{n=1}^{N}(a_n-\hat{a}_n)(y_n-t_n)$$

$$=\frac{1}{2}\|\underline{w}\|^2-\sum_{n=1}^{N}(a_n+\hat{a}_n)\varepsilon-\sum_{n=1}^{N}(a_n-\hat{a}_n)(\underline{w}^T\phi(x_n)+b-t_n)$$

$$=\frac{1}{2}\|\underline{w}\|^2-\sum_{n=1}^{N}(a_n+\hat{a}_n)\varepsilon-\sum_{n=1}^{N}(a_n-\hat{a}_n)(\underline{w}^T\phi(x_n)+b)$$
$$+\sum_{n=1}^{N}(a_n-\hat{a}_n)t_n$$

(7.58) $$= \frac{1}{2}\|\underline{w}\|^2 - \sum_{n=1}^{N}(a_n+\hat{a}_n)\epsilon - \sum_{n=1}^{N}(a_n-\hat{a}_n)(\underline{w}^T\phi(x_n)) + \sum_{n=1}^{N}(a_n-\hat{a}_n)t_n$$

(7.57) $$= \frac{1}{2}\|\underline{w}\|^2 - \sum_{n=1}^{N}(a_n+\hat{a}_n)\epsilon - \|\underline{w}\|^2 + \sum_{n=1}^{N}(a_n-\hat{a}_n)t_n$$

$$= -\frac{1}{2}\|\underline{w}\|^2 - \sum_{n=1}^{N}(a_n+\hat{a}_n)\epsilon + \sum_{n=1}^{N}(a_n-\hat{a}_n)t_n$$

which is equivalent to (7.61), employing a formulation with the kernel matrix, when we invoke (7.57).

Relationship between Linear and Kernel SVM

We will show that kernel SVM and linear SVM are equivalent when a linear kernel is used.

To this end, we first consider the linear version of the SVM problem (see equations (15)-(17) on slideshow 20),

$$\arg\min_{\underline{w}, b} \frac{1}{2} \|\underline{w}\|^2 + C \sum_{n=1}^{N} \xi_n,$$

$$\text{s.t.} \quad t_n (\underline{w}^T \underline{x}_n + b) \geq 1 - \xi_n, \; n = 1, \dots, N,$$

$$\xi_n \geq 0, \; n = 1, \dots, N.$$

In analogy to equation (18), we introduce the Lagrangian

$$L(\underline{w}, b, \underline{\alpha}) = \frac{1}{2} \|\underline{w}\|^2 + C \sum_{n=1}^{N} \xi_n - \sum_{n=1}^{N} \alpha_n \left[t_n (\underline{w}^T \underline{x}_n + b) - 1 + \xi_n \right] - \sum_{n=1}^{N} \mu_n \xi_n$$

with $\underline{\alpha} = [\alpha_1, \dots, \alpha_n]$ as the Lagrangian multipliers and (relatively) straightforward derivatives

$$\frac{\partial L}{\partial \underline{w}} = 0 \Rightarrow \underline{w} = \sum_{n=1}^{N} \alpha_n t_n x_n,$$

$$\frac{\partial L}{\partial b} = 0 \Rightarrow \sum_{n=1}^{N} \alpha_n t_n = 0, \qquad (*)$$

$$\frac{\partial L}{\partial \xi_n} = 0 \Rightarrow C = \alpha_n + \mu_n, \quad n = 1, \dots, N. \quad (\ast\ast)$$

Substituting into the Lagrangian yields

$$\mathcal{L} = \frac{1}{2} \left\| \sum_{n=1}^{N} \alpha_n t_n x_n \right\|^2 + C \sum_{n=1}^{N} \xi_n$$

$$- \sum_{n=1}^{N} \alpha_n \left[t_n \left(\left(\sum_{m=1}^{N} \alpha_m t_m x_m \right)^T x_n + b \right) - 1 + \xi_n \right]$$

$$- \sum_{n=1}^{N} \mu_n \xi_n$$

This is a bit of a complex equation, but it simplifies quite a lot. Note first that

$$\frac{1}{2} \left\| \sum_{n=1}^{N} \alpha_n t_n x_n \right\|^2 = \frac{1}{2} \left(\sum_{n=1}^{N} \alpha_n t_n x_n \right)^T \left(\sum_{n=1}^{N} \alpha_n t_n x_n \right)$$

$$= \frac{1}{2} \sum_{m=1}^{N} \sum_{n=1}^{N} \alpha_m \alpha_n t_m t_n x_m^T x_n,$$

Furthermore,

$$- \sum_{n=1}^{N} \alpha_n \left[t_n \left(\left(\sum_{m=1}^{N} \alpha_m t_m x_m \right)^T x_n + b \right) - 1 + \xi_n \right]$$

equal to zero by $(\ast)$

$$= - \sum_{m=1}^{N} \sum_{n=1}^{N} \alpha_m \alpha_n t_m t_n x_m^T x_n - b \cdot \left(\sum_{n=1} \alpha_n t_n \right)$$

$$+ \sum_{n=1}^{N} \alpha_n - \sum_{n=1}^{N} \alpha_n \xi_n$$

Finally, we have that

$$C\sum_{n=1}^{N} \xi_n = \sum_{n=1}^{N} C\ \xi_n$$

$$= \sum_{n=1}^{N} (\alpha_n + \mu_n)\ \xi_n$$

Using (***)

$$= \left(\sum_{n=1}^{N} \alpha_n \xi_n\right) + \left(\sum_{n=1}^{N} \mu_n \xi_n\right)$$

Thus, the Lagrangian reduces

$$L = \frac{1}{2}\sum_{m=1}^{N}\sum_{n=1}^{N} \alpha_m \alpha_n t_m t_n x_m^T x_n$$

$$+ \sum_{n=1}^{N} \alpha_n \xi_n + \sum_{n=1}^{N} \mu_n \xi_n$$

$$- \sum_{m=1}^{N}\sum_{n=1}^{N} \alpha_m \alpha_n t_m t_n x_m^T x_n + \sum_{n=1}^{N} \alpha_n - \sum_{n=1}^{N} \alpha_n \xi_n$$

$$- \sum_{n=1}^{N} \mu_n \xi_n$$

$$= \sum_{n=1}^{N} - \frac{1}{2}\sum_{n=1}^{N}\sum_{m=1}^{N} \alpha_n \alpha_m t_n t_m x_n^T x_m$$

But this is exactly the dual Lagrangian we obtained in the kernel case (see equation (22) slideshow 20) if we use a linear kernel.